NorthStar Infrastructure Group – AI Engineer Case Assignment

Agentic Operations Programme · Candidate Case Exercise

Case Introduction

NorthStar Infrastructure Group (NIG) is a fictional global engineering, construction, and operations company with approximately 50,000 employees across North America, Europe, the Middle East, and Asia-Pacific.

Over the last decade, NIG has grown through acquisitions and regional expansion. As a result, many business functions operate differently across regions, business units, and projects. Finance, HR, Procurement, Legal, Sales, and Operations teams often rely on different systems, approval processes, reporting structures, and local workarounds.

Leadership believes a significant portion of operational work can be automated or augmented through AI. However, previous transformation programmes have struggled due to fragmented data, inconsistent processes, regulatory requirements, local business needs, and resistance to change.

NIG has approved a multi-year programme to modernise internal operations through an AI-enabled operating platform. The objective is to improve efficiency, quality, compliance, and scalability while maintaining appropriate human oversight and accountability.

The platform is expected to support operational workflows across the following clusters:

- FIN · Finance
- HR · People & HR
- LEG · Legal
- IT · IT & Security
- PRO · Procurement
- COM · Comms & Growth
- KNW · Knowledge & Ops
- ANL · Analytics & Strategy
- GWT · Sales & Marketing

The company cannot transform everything at once. Budget, engineering capacity, regulatory obligations, stakeholder alignment, and business continuity are all constraints.

Candidates should assume the information provided is intentionally incomplete. We are not looking for perfect assumptions or deep domain expertise. We are interested in how you identify unknowns, prioritise problems, make trade-offs, and communicate decisions under uncertainty.

Case Introduction – AI Engineer

NIG wants to rapidly identify and validate high-value operational use cases. Leadership has approved experimentation, but solutions must demonstrate clear business value and realistic implementation paths — the organisation has seen too many impressive demos that never survived contact with real operations.

As AI Engineer, you are expected to move quickly, make practical technology decisions, and demonstrate working software rather than theoretical concepts. You will work within NIG's Finance cluster (FIN), build a solution for a specific Finance workflow, and defend your engineering judgement.

Certain over-used workflows are deliberately excluded to force creativity: you may not build invoice processing, CV screening, contract review, ticket routing, knowledge chatbots, or lead qualification.

Submission Expectations

Candidates should assume that the context provided is intentionally incomplete. Where information is missing, state reasonable assumptions explicitly and explain how those assumptions influence your proposed approach.

We value clarity, prioritisation, practical thinking, and communication over polished production-ready deliverables. Strong candidates may reach very different conclusions and still succeed; we are evaluating the quality of reasoning, not a single correct answer.

Your submission consists of a working prototype, a recorded demo video (5–10 minutes), a technical design document, and an executive summary. Mocked or simulated data is acceptable and expected.

We are evaluating engineering judgement, build-vs-buy reasoning, speed of delivery, and the realism of your path to production — not polish. Use modern AI tooling and workflows to validate ideas quickly.

- Working prototype with source code or hosted access
- Recorded demo video (5–10 minutes) showing at least one core workflow end-to-end
- Technical design document and executive summary, submitted prior to the interview session
- Live session: ~30 minutes presentation and demo, ~30 minutes technical and business Q&A;

Task 1 — Build a Finance Operational Solution

NIG's Finance function (FIN) spans budget planning, project cost control, month-end close, financial reporting, variance analysis, forecasting, and compliance. These workflows consume significant time from skilled finance professionals across regions and business units. Leadership does not want another generic assistant — it wants evidence that a specific, painful Finance workflow can be meaningfully automated or augmented, with a credible story about who uses it and why they would trust it.

Budget for experimentation is real but finite; a prototype that requires six months of data engineering before it works is, for this purpose, a failed prototype.

Select a workflow from the FIN · Finance cluster. You may NOT build: invoice processing, CV screening, contract review, ticket routing, knowledge chatbots, or lead qualification.

Build a working prototype and demonstrate it through a recorded video. Define the Finance user, the current pain, and the future-state workflow. Mocked or simulated data is acceptable — but it should be realistic enough to expose the hard parts of the problem rather than hide them.

Task 2 — Technical Design Review

NIG's engineering leadership has learned to distrust demos that cannot explain themselves. Every prototype that requests further funding goes through a design review where the choices — not the output — are interrogated. Vendor components are welcome where they make sense, but every buy decision creates a dependency the platform team must live with.

Document the architecture, tooling choices, models, integrations, and evaluation strategy behind your prototype. Explain what you bought versus built and why, what you would change with more time, and how you know the system works — including how you measured quality and the failure cases you observed.

Task 3 — Production Readiness

The gap between a prototype and a production capability at NIG is wide: real data access across fragmented systems, identity and permissions, multiple languages and jurisdictions, support and on-call models, cost at scale, and the trust of users who did not ask for the tool. Several promising NIG pilots have died precisely in this gap.

Describe the work required to scale your Finance solution to production usage across multiple teams, projects, and countries. Be specific about data, security, reliability, cost, and operational ownership. Estimate the effort honestly — leadership values a credible 'this needs two quarters and these three things' over an optimistic hand-wave.

Task 4 — Executive Summary and Demo Video

Funding decisions at NIG are made by executives who will spend five minutes on your material. They have been burned by AI hype and respond to clear value, honest risk, and realistic cost. Your demo video may be forwarded without you in the room to defend it.

Create a concise executive summary covering business value, expected ROI, risks, costs, and rollout approach, accompanied by your recorded demonstration. The summary must stand alone: assume the reader has not seen any of your other materials.

Task 5 — Assumptions & Kill Test

NIG's portfolio approach to AI experimentation depends on killing weak solutions quickly so resources flow to strong ones. Engineers who cannot articulate how their own solution should be judged — and ended — are a portfolio risk.

Identify the three assumptions your solution depends on most heavily. For each, explain why it matters, how you would validate it, and the consequences if it proves false.

Then define three metrics that would convince you the solution should be shut down or fundamentally redesigned, including the measurement window and thresholds.